

# Vignesh Balaji Ravichandran

475-317-4536 | [VigneshBalaji.Ravichandran@quinnipiac.edu](mailto:VigneshBalaji.Ravichandran@quinnipiac.edu) | [linkedin.com/in/vignesh-balaji-95830516b/](https://www.linkedin.com/in/vignesh-balaji-95830516b/) | <http://www.github.com/baloog>

---

## EDUCATION

### Quinnipiac University,

Master of Science in Computer Science,

GPA – 4.0/4.0

Hamden, CT, USA

Aug 2024 – May 2026

### SCSVMV University,

Bachelor of Engineering in Computer Science,

GPA – 8.6/9.0

Kanchipuram, TN, India

Aug 2016 – May 2020

---

## EXPERIENCE

### Full Stack Developer

#### Goaira Technologies Pvt Limited | Nov 2022 – Apr 2024 | Chennai, India

- Developed and optimized full-stack applications, improving system performance by ~25% and reducing response latency
- Reduced production defects by ~30% through rigorous peer code reviews and adherence to SDLC best practices
- Managed Agile workflows using **Jira**, increasing sprint delivery efficiency by ~20% and improving task traceability
- Refactored legacy codebases, reducing technical debt by ~35% and improving maintainability and scalability
- Built and maintained REST APIs, enabling seamless integration across systems and improving data flow reliability

---

### Full Stack WordPress Intern

#### Ijona Technologies Pvt Ltd | May 2021 – Feb 2022 | Bangalore, India

- Migrated 15+ **websites** from legacy systems with <5% **downtime**, ensuring smooth transition and continuity
- Managed hosting and backend operations, improving system uptime and reliability by ~20%
- Implemented automated backup systems, reducing data loss risk and improving recovery time by ~40%
- Optimized server configurations and storage management, improving overall system efficiency

---

## Projects

### Longitudinal Chest X-ray Progression Analysis | PyTorch, ResNet-18, OpenCV | Sept 2025 – Present

- Developed a ResNet-18 based multi-label classifier to detect 8 thoracic diseases from chest X-rays
- Designed a system to analyze disease progression over time at the patient level (improved, worsened, stable)
- Achieved 0.79 macro AUROC and ~82% classification accuracy across multiple diseases
- Improved performance by ~12% through preprocessing and tuning
- Integrated Grad-CAM visualizations with bounding box generation, improving interpretability by ~30%

---

### Heart Disease Prediction System | Scikit-learn, Pandas, NumPy, Python | Jan 2026 – Present

- Built a multi-class classification model using Random Forest and SMOTE to handle class imbalance
- Predicted disease severity levels: no disease, mild, and moderate-to-severe cases
- Achieved ~85% accuracy and ~0.83 F1-score on balanced dataset
- Improved performance by ~15% through feature engineering and preprocessing
- Reduced class imbalance impact by ~40% using SMOTE

---

## Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript

Frameworks/Libraries: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, React, Node.js, Flask, FastAPI

Tools: Git, Docker, Jira, Google Cloud Platform, VS Code, PyCharm

Concepts: SDLC, Agile, Machine Learning, REST APIs, Full-Stack Development